INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 5: Discussion

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5 Discussion

Chapter 4 reported the measurements taken on Arbitrum One. Here each research question and each objective of Chapter 1 is held to the criterion stated there, and every interpretation points to the table of Chapter 4 it rests on. Unless noted otherwise, same-window statements concern the matched cohort of contracts and holdout statements the spender cohorts.

5.1 Findings by research question

5.1.1 RQ1: Reputation structure and rank divergence

RQ1 asks whether EndorseRank and AWP order the same contracts in different ways. They do: the interval of their agreement excludes both zero and one (Table 4.5). The divergence does not come from ties or from the rule for missing contracts: EndorseRank leaves few contract pairs tied, and keeping the contracts missing from a graph as isolated nodes leaves the agreement unchanged (Section 4.4).

The divergence follows from how the two graphs are built. The endorsement graph records the latest approvals and has fewer than half of the edges of the transfer graph, which records every transfer (T. Do et al., 2023), and the two methods restart differently. PageRank places stationary mass according to the edges it is given (Langville & Meyer, 2006; Page et al., 1999), so a contract that receives many transfers from active senders but few approvals ranks high under AWP and modestly under EndorseRank, and the reverse holds for a contract that many owners approve without being a transfer hub. H1 is supported, largely by construction, since two walks over different graphs with different restarts were not expected to agree.

5.1.2 RQ2: Same-window alignment

RQ2 asks how closely each score agrees with contemporaneous allowance and transfer checks. Each score follows the edges it walks on: the walks over transfers agree most with the transfer family and much less with the allowance family, the walks over allowances agree most with the allowance family, and every interval on a score's own family excludes zero (Table 4.3).

Circularity bounds H2. Every proxy in a family is computed from the same events as the matching graph, so the agreement is an intended-construct check in the sense of Cronbach and Meehl (1955) and not an external test. Part of it is built in: AWP, whose edges weigh each transfer by its dollar value, follows the summed value of inbound transfers almost as closely as their number (Table 8.7). Ties do not bound H2 either: the isolated-node rule changes no coefficient (Table 4.5). In the narrow sense of agreement with the checks of a score's own edges, H2 is supported, largely by construction. None of these comparisons was fixed in advance, and the paired differences between EndorseRank and AWP, which carry no claim, are in Appendix 8.5.

5.1.3 RQ3: Computational cost

RQ3 asks what each score costs to compute on the same sample. EndorseRank is the cheapest score in time, memory and iterations, and the two hybrids, which walk the transfer layer, take longer than AWP (Table 4.1). Each iteration costs time in $|E|$ and in $|V|$ (Langville & Meyer, 2006), and AWP needs more iterations than EndorseRank, so its time ratio lies just below the edge ratio at $d = 0.85$ and moves with d (Section 4.2): the cost grows with the size of each graph and with the iterations its walk needs, which the benchmark does not separate.

The scaling tables qualify this result more than they extend it (Table 4.2). Each stage is a subsample of the cohort that keeps the edges incident to its contracts, so the stages never test a graph larger than the cohort graph (Section 4.2.1). The damping and value-slope checks leave the alignment pattern as it was (Tables 4.6 and 4.7). H3 is supported on the matched cohort and on the subsamples from 10,000 contracts on, largely by construction, since both scores run the same solver; on the

smaller stages only EndorseRank's lead in speed holds. H1–H3 are thus descriptive checks of the instruments; the tests whose outcome was open are RQ4, RQ5 and Section 5.1.6.

5.1.4 RQ4: Temporal holdout of future new approvals

RQ4 has two clauses. The first asks whether scores frozen at the end of March 2026 predict new owner–spender approvals and new transfer senders on the spender cohort over the following three months. Every score predicts both labels, and the edge type decides how well (Table 4.10). On new approval pairs the interior mixtures and the in-approve degree lead, followed by EndorseRank and the allowance layer, while the scores built on transfers alone stay lower, mainly because they score zero the one spender in six without a transfer edge, and these gained new approval pairs slightly more often. On the other spenders AWP ranks this label above EndorseRank and the in-approve degree in W0, without a paired contrast, and above C-PR in both windows (Tables 4.12 and 4.20; restrictions chosen after the labels were known). On new senders AWP leads, ahead of the transfer in-degree, and the scores built on allowances alone rank lowest. W1 repeats this pattern among the same scores (Tables 4.14 and 4.16). The evidence concerns spenders; Section 5.1.6 turns to the traders.

The second clause asks whether any PageRank beats the raw degree it smooths, and the two walks answer it differently (Tables 4.11 and 4.16). AWP ranks later new senders above the transfer in-degree in both windows, with intervals above zero, and in W0 its lead holds on the spenders that had received a transfer before the freeze, so the spenders it scores zero do not create it (Table 4.12). EndorseRank, with the uniform restarts of the analysis plan, ranks new approval pairs below the in-approve degree in both windows. Its variant with AWP's activity restarts ranks them above the degree in both windows, and the transfer layer of C-PR, walked with uniform restarts, stays slightly below the transfer in-degree, without a paired contrast (Section 4.6, Table 4.14). This suggests that the restart rule, not the edge type, decides whether a walk beats its count, a hypothesis for a registered test and not a finding. Known results lead one to expect the count to be hard to beat: in growing networks new links attach preferentially to nodes that already have many (Barabási & Al-

bert, 1999; Newman, 2001), which makes degree a standard link-prediction baseline (Liben-Nowell & Kleinberg, 2007), also on Ethereum transaction networks (D. Lin et al., 2022), and where degree correlations are weak, as on the Web, PageRank is on average proportional to in-degree (Fortunato et al., 2008). For most spenders EndorseRank is a reweighted approval count (Section 5.3.2). This does not undo the first clause, since the counts are statistics of the same edges. All of these contrasts were computed after the labels were known.

The labels lie outside the window but belong to the same constructs as the scores, and neither contains any element of profit or repayment. RQ4 therefore shows the temporal predictive validity of each construct, with the raw degree as the baseline that a PageRank must beat, and leaves credit risk aside; the liquidation label of W1 (Section 3.7.2), on which Section 5.1.6 compares the edge types, comes closest to it.

5.1.5 RQ5: Coupling the two edge types

For C-PR, the coupling the plan fixed, both halves of RQ5 get a negative answer, each under its own rule, applied here, like R1–R3 in Section 5.1.6, to the revised scores as a re-analysis; the registered analysis reached the same verdicts (Appendix 8.6). The second half asks whether one walk over both layers ranks contracts better than either layer walked alone. Under rule A, fixed in the analysis plan, it does not. On W0 C-PR is no worse than the allowance layer on new approval pairs but falls behind the transfer layer on new senders (Table 4.11), and in the same window the secondary rule fails on all three families (Section 4.3.1). The first half asks whether the walk improves on a walk over the transfers alone. An exploratory comparison on W0 said that it did on new approval pairs, but it was made after the labels were known. Rule B, registered before the study extracted the July–September logs, answers no: C-PR again beat the transfer layer on new approval pairs, but non-inferiority was shown neither on new senders nor on the traders' liquidation-free close rate, and the rule needed all three conditions (Section 4.6.2). In the first sensitivity analysis, with AWP as the comparator, F1 holds and F2 and F3 fail: C-PR ranks new approval pairs higher and new senders lower, and the close rate does not separate the two (Table 4.15); W0, computed after the

labels were known, agrees (Table 4.11). The lead comes from the spenders AWP scores zero, and on the others AWP ranks both labels higher (Section 4.6.4). Against EndorseRank, C-PR is above on new approval pairs in both windows, also post hoc (Table 4.16).

The ablation and the layer weights explain both halves. S-PR keeps the transfer walk but restarts it from the allowance walk; it reproduces the transfer walk on the flow label and loses most of the endorsement signal, since a teleportation vector derived from endorsements changes little when transfer edges decide where the walk goes, and since S-PR scores only the transfer-layer nodes, part of that loss is coverage. In C-PR the allowance edges carry part of the walk, so the allowance signal survives on the spender holdout: every interior mixture is slightly above the allowance layer on new approval pairs and below the transfer layer on new senders (Tables 4.10 and 4.11). Unlike AWP, which bounds the weight of each transfer and restarts at active senders, C-PR's layers carry dollar amounts and restart uniformly; walked alone, its transfer layer already ranks new senders below AWP in both windows, by about three fifths of C-PR's gap, so most of that gap comes from the weights and restarts, not from the coupling. C-AWP, the same coupling of AWP and EndorseRank with activity restarts, loses much less on new senders and ranks new approval pairs above AWP, also among the spenders AWP orders; it still fails the structure of rule B against AWP, and as it was defined after the labels were known, it is the coupling to register on a later window, not a finding (Section 4.6.4).

Both halves count as results. The rule for the second half was fixed before any coupled score existed, so λ could not have been tuned to it, although the revised C-PR's dollar weights and allowance cap were set after the registered results were known. It limits an obvious next step: the plan's convex mixture of the two edge types at the transition level gained on new approval pairs over the allowance layer and lost on new senders against the transfer layer. On both labels at once, which neither rule judged, C-PR leads AWP on new counterparties in both windows but trails it on the mean of the two coefficients in W0 and is slightly ahead in W1 (Table 4.20, post hoc). W1 establishes something narrower: adding allowance edges to C-PR's transfer walk ranks new approval pairs better on new data, partly through the spenders the walk scores zero, at a loss on new senders larger than the

margin of the registration; whether C-PR beats its transfer layer on the other spenders was not computed.

5.1.6 Allowance against transfer on outcomes built from neither edge type

Unlike the spender labels of RQ4 and RQ5, which favor the scores built on their own edge type, the GMX trader outcomes are built from neither edge type and were measured after the scores were frozen. The comparison of the two edge types on them (Section 3.7.3) adds no research question and is reported as Claim C6 (Section 4.6.3).

This comparison bears on counterparty risk, the use that motivated the thesis (Chapter 1). For a protocol that extends leverage or credit to a contract account, a counterparty that is often force-liquidated is a riskier one, and forced liquidation is the nearest outcome to default in these data. On later liquidation avoidance in W1, the primary window, the difference P between the approval count and the transfer count at the freeze has a Bonferroni 97.5% interval that includes zero (Table 4.18), as does the interval stratified by the number of closes, so R1 is not met: there is no count-level lead of allowance over transfer on later liquidation avoidance. In W0, the secondary window, P was positive. The transfer count also ranks this outcome (Table 4.14), and the transfer scores rank as well the outcomes that grow with activity, the number of profitable closes and the realized gain (Table 4.17): they record how much a contract trades.

Four limits bound this reading. First, R2 is not met: the like-for-like PageRank pair, EndorseRank with AWP's restarts against AWP, is not resolved in either window, and EndorseRank again falls below the approval count it smooths (Table 4.18). Second, the comparison rests on a few dozen approval-receiving traders per window, all of whom avoided liquidation, against about three in four of the others (Table 4.19), and every interval is wide. Third, every trader is a contract, so account type cannot explain the pattern, but the role of a contract may: nearly all recipients were approved by one contract (Section 3.7.3), so they form one cluster, and they close more positions, which fits a family of managed or pooled trading contracts. Fourth, R3 applies in W1: the same signal is lower on the share of closes that were profitable, so the construct it ranks is forced-liquidation

avoidance, not trading success; its lower non-loss share is partly built in, because the pipeline counts a liquidated close as not a loss.

5.2 Achievement of the research objectives

Chapter 1 attached a measurement and a criterion to each of its five objectives. Table 5.1 checks them against the evidence.

Five findings outside the objectives also shape the conclusions. Three limit the claims: the value slopes chosen in the revision were checked only in the same window (Section 5.8.1); on the trader cohorts no score ranks the share of profitable closes with an interval above zero (Section 4.6.1); and EndorseRank's point estimates on both spender labels lie slightly above those of the allowance layer of C-PR, without a paired contrast, so how approvals are weighted may matter (Section 5.8.1). Two add to the claims: every trader contract that had received an approval, nearly all from one contract, avoided liquidation in both windows, in small samples (Section 5.1.6), and for most cohort spenders, those of depth one, EndorseRank reduces to an owner-normalized in-strength (Section 5.3.2).

5.3 Theoretical implications

The results bear on three areas of theory: propagation semantics on financial graphs, the construct validity of on-chain scores, and the economics of trust under pseudonymity.

5.3.1 Propagation semantics beyond transfers

PageRank was first formulated for hyperlinks, read as a citation network (Page et al., 1999). H.-D. Do and Do (2023) carried this reading over to Ethereum transactions, and AWP (T. Do et al., 2023) refined it with value and age weights and restarts at active senders. EndorseRank applies it to ERC-20 allowances, where an owner who lets a spender act for them issues a directed endorsement of spending authority (Vogelsteller & Buterin, 2015). The RQ1 divergence shows that changing what the edges mean, and where the walk restarts, changes the ranking even on the same nodes, so graph-

Table 5.1: Research objectives, the criterion stated in Chapter 1, and the outcome.

Obj.	Criterion	Evidence	Outcome
O1	95% interval of inter-method τ contains neither 0 nor 1	Table 4.5, last row: the interval $[0.163, 0.185]$ lies strictly inside $(0, 1)$	Met (largely by construction)
O2	A score agrees with a construct when its interval on that family excludes 0; agreement with the construct of a score's own edges is an intended-construct check	Table 4.3: every walk over transfers alone on the transfer family (0.572 $[0.564, 0.580]$ to 0.659 $[0.653, 0.666]$) and both walks over allowances on the allowance family (0.585 $[0.578, 0.592]$ and 0.597 $[0.590, 0.604]$); every interval excludes 0 except EndorseRank's on Sybil stability; the proxies come from the same events as the graphs	Met (largely by construction)
O3	Runtime ratio given alongside the edge ratio, under a single fixed protocol	Table 4.1: mean, SD, memory, iterations, $\lvert V\rvert$ and $\lvert E\rvert$ on the full graphs; Table 4.2: mean time, $\lvert V\rvert$ and $\lvert E\rvert$ per stage; time ratio 2.17 beside an edge ratio of 2.23, and 2.55 to 1.95 across damping values (Table 4.6); one solver for all scores	Met (largely by construction)
O4	A score predicts a label when its interval excludes 0; a PageRank improves on its degree only when the interval of the paired difference lies above 0	Tables 4.10, 4.14 and 4.16: every score predicts both labels in W0 and W1. Tables 4.11 and 4.16: AWP minus transfer in-degree $+0.043$ $[0.038, 0.048]$ (W0) and $+0.022$ $[0.017, 0.027]$ (W1); EndorseRank minus in-approve degree -0.019 $[-0.026, -0.011]$ (W0) and -0.034 $[-0.038, -0.029]$ (W1); EndorseRank with activity restarts minus in-approve degree $+0.059$ $[0.050, 0.067]$ (W0) and $+0.027$ $[0.021, 0.034]$ (W1); all computed after the labels were known	First clause met; second met for AWP and the activity variant, not for EndorseRank
O5	(a) Rule A, fixed in the plan: on W0, C-PR no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new senders; in the same window, within the leading layer's interval on the transfer and allowance families and above both layers on Sybil stability. (b) Rule B, registered: on W1, C-PR beats C-PR ($\lambda = 0$) on new approval pairs and is not worse by more than 0.01 on new senders and on the traders' liquidation-free close rate	Revised scores, re-analysis. (a) Table 4.11: no worse holds ($+0.029$ $[0.022, 0.036]$), better fails (-0.034 $[-0.040, -0.029]$). Same window: all three parts fail (Section 4.3.1). (b) Table 4.15: new approval pairs $+0.110$ $[0.098, 0.122]$ holds; new senders -0.028 $[-0.033, -0.023]$ and liquidation-free close rate $+0.000$ $[-0.014, 0.010]$ fail. Sensitivity 1, AWP as comparator: $+0.081$ $[0.069, 0.095]$ holds, -0.067 $[-0.072, -0.061]$ and $+0.021$ $[-0.019, 0.060]$ fail; post hoc comparisons in Table 4.20	(a), (b) not met, as in the registered analysis (Table 8.12)

based reputation is a family of methods, each valid relative to the relationships its edges encode. Peer-to-peer and web trust systems reached the same conclusion from the other side: in EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) the seed set and the edge definitions largely decide the outcome.

The holdouts add a qualification that the propagation metaphor hides. Changing the edge type changed what the score could anticipate: on new approval pairs the scores built on allowance edges led, mainly through the spenders that walks over transfers score zero, and on new senders the walks over transfers led, in both windows (Section 5.1.4). Whether the walk added to the count appears to have depended on the restart rule, not on the side: AWP and EndorseRank with activity restarts beat their counts, while EndorseRank and the transfer layer of C-PR, both with uniform restarts, did not. On these data the choice of edge decides which spenders a score can order, and the restart rule whether the walk improves on the count. When the two layers share one walk, the allowance signal survives where allowance edges carry the walk and is mostly lost where they only seed its restarts (Section 5.1.5).

5.3.2 EndorseRank on depth-one spenders

The allowance graphs keep only the approvals of the five tokens that a cohort contract grants or receives (Section 3.6), so most nodes are pure owners, with out-edges only, or pure spenders, with in-edges only. A cohort contract can be both, so the graph also has paths of more than one step. For a spender whose owners receive no approvals, EndorseRank has a closed form.

Proposition. Let PageRank be computed by Equation (3.1) with damping d and uniform restarts, so that dangling mass is also spread uniformly. Then every node without in-edges has the same score c, and every node v whose in-neighbors all lack in-edges has

$$r_v = c\,\big(1 + d\,\delta(v)\big), \qquad \delta(v) = \sum_u \frac{w(u,v)}{o(u)}, \tag{5.1}$$

where $\delta(v)$ is the owner-normalized in-strength of v: each owner's endorsement is divided over all

the spenders it approves.

Derivation. At the fixed point, $r_v = (1 - d + d\, r_D)/|V| + d \sum_u r_u\, w(u,v)/o(u)$, where r_D is the mass held by the dangling nodes. A node without in-edges receives only the first term, so every such node has the score $c = (1-d+d\, r_D)/|V|$, the constant c of Section 5.6.3. If every in-neighbor u of v is such a node, substituting $r_u = c$ gives Equation (5.1).

On the graph at T_{obs}, 12,039 of the 14,406 cohort spenders have depth one, and EndorseRank as computed in this thesis lies within 1% of Equation (5.1) for every one of them; of the deeper spenders, which have an owner that itself receives approvals, 26.4% do. Over all spenders the share is 87.9%, and over the W0 spenders at t_1 it is 73.7% (scripts `run_same_window.py` and `run_holdout.py`). Two consequences follow. First, for a depth-one spender EndorseRank only reweights the approval count, counting an owner that approves many spenders for less at each; besides the time and value weights, that normalization is all that separates it from the count, which ranks new approval pairs better (Section 5.1.4). For these spenders the negative answer to the second clause of RQ4 for EndorseRank thus has an analytic basis: there was no propagated information for the walk to add. With activity restarts, dangling mass follows the restarts, so for a depth-one spender the variant is a one-step sum over its owners weighted by their restart mass: its gain over the count comes from how owners are weighted, not from longer paths. Second, the graph does have deeper paths, and for most deeper spenders the score departs from the closed form, so multi-hop propagation is present in this graph, but over all spenders EndorseRank did not beat the raw count out of window, and the deeper spenders were not tested apart. The ego network cuts longer paths (Section 5.8.3), so testing propagation further would need a deeper sample, such as the two-hop neighborhood of the cohort or a chain-wide allowance graph, or restarts seeded at vetted contracts.

5.3.3 Multi-construct domain alignment

The checks were designed to separate transfer centrality from allowance delegation, and each score aligns with the construct of the edges it walks on, the mixtures falling between their layers. Classical construct-validity theory would predict this pattern for instruments that measure different constructs

(Cronbach & Meehl, 1955), and the integrated view of Messick (1995) goes further, counting the consequences of using a score as part of its validity. The same theory says what such a pattern cannot settle: when two instruments measure different constructs, the one that comes out ahead on a check is the one built for that check, so ordering the instruments by their coefficients does not show which is the better measure of reputation. Outcomes built from neither edge type are the exception, since neither instrument was built for them, which is why the thesis compares the edge types there (Section 5.1.6); even there, which edge type leads depends on the outcome.

5.3.4 Reputation under cheap pseudonyms

Identities are cheap to create on public ledgers (Nakamoto, 2008), and under Ethereum's account model a new address needs no registration step at all (Wood, 2025). Any reputation system that rewards inbound transfers or allowances is therefore exposed to Sybil strategies, in which an attacker creates edges among addresses it controls (Douceur, 2002). Section 5.6 formalizes the cost of attacking an allowance graph, where a closed farm of addresses that approve one another needs, unlike transfer wash trading, no token balance and pays little beyond gas.

5.4 Practical implications and deployment scenarios

The results support fitting a score to each decision over recommending a single one, which also lowers the risk of using a flow score to decide authorization. The degree contrasts cited below were computed after the labels were known.

5.4.1 Transfer-hub and flow monitoring

A protocol that describes current flow (transfer-hub search by market makers and liquidity providers, routing analysis, or transfer-based compliance heuristics) can read it from in-degree and in-value directly. AWP (T. Do et al., 2023) agrees with present inflow (Table 4.3) and out of window ranked later new senders better than any other score, the raw transfer in-degree included, in both windows (Tables 4.10 and 4.16), so a protocol that wants to anticipate flow has reason to use it over the count.

5.4.2 Authorization-aware and low-latency refresh

Where explicit spending authorization matters most (token vaults, allowance-gated integrations, or risk engines that treat approvals as acts of trust), protocols have reason to use allowance edges. The cheapest form is the spender's in-approve degree, which needs no solver and predicted new approvals better than EndorseRank in both windows (Tables 4.11 and 4.16). EndorseRank, its propagated version, agrees most with the allowance checks (Table 4.3) and solves faster than AWP (Table 4.1), but for most spenders it reflects the endorsing owners only through how many spenders each approves (Section 5.3.2), a property not shown to improve any label, and under uniform teleportation it is open to cheap inflation by closed farms (Section 5.6). The variant with activity restarts ranked new approval pairs above the count after the labels were known, so it is the propagated allowance score to test.

5.4.3 Evaluation design and combined use

An operator can run EndorseRank for continuous authorization-aware monitoring and AWP for periodic transfer-hub audits, at the cost of both solves. Folding the two edge types into one walk as C-PR does, with uniform restarts and dollar-weighted layers, traded one label for the other on these data: against AWP it ranks new approval pairs higher, through the spenders AWP scores zero, and new senders lower (Section 5.1.5), and one C-PR solve takes longer than the EndorseRank and AWP solves together (Table 4.1). C-AWP, a coupling of AWP and EndorseRank with activity restarts, loses much less on new senders, but it was defined after the labels were known and still fails the structure of rule B against AWP (Section 4.6.4). Until a coupling is registered and tested on a later window, an operator free to choose has no tested reason to replace the two scores with one walk.

5.4.4 Industry adoption scenarios

The scenarios below say for whom each score is useful and for whom it is not.

- DEX operators. Fee tiers or monitoring priority by user need a score for users, who are mostly externally owned accounts and are not scored here (Section 3.2); the scores can instead rank the routers, vaults and pools a DEX integrates with.
- Lending and margin protocols. Lending markets want to put more of their capital to use without cutting collateral (Bartoletti et al., 2021). When the counterparty is a contract, their screening question is whether it will be force-liquidated, and on that outcome the approval count led the transfer count in W0 but not in the primary window, on a few dozen contract traders per window (Section 5.1.6). A screen built on it would need a registered test on a larger sample; the scores are not tools for predicting default.
- Wallet and security tooling. A badge that shows how many owners have approved a spender contract, or its EndorseRank, tells users and integrators how widely it is authorized (Vogelsteller & Buterin, 2015); it is neither a security audit nor a guarantee against malicious contracts.
- Airdrop and governance programs. Sybil clusters routinely abuse token distributions and governance eligibility (Luo et al., 2025). EndorseRank as defined here offers little protection below the top of its ranking, since about two dozen farm addresses lift a contract into the cohort's top tenth (Section 5.6). Programs can require endorsements from owners with independent histories, discount mass that circulates inside closed clusters, or send teleportation and dangling mass only to a vetted seed set; none of these stops an attacker who buys endorsements from honest owners.

In none of these scenarios does reputation replace collateral.

5.4.5 Standardized scores as shared infrastructure

A score shared by several parties, such as exchanges, funds and compliance teams, is useful only if each of them can recompute it and see how it might be manipulated. EndorseRank and AWP can be recomputed with the open pipeline from public logs and a daily price series, on any ERC-20

chain with allowance logs (Appendix 8.1), and Sections 5.6 and 5.7 document how they can be manipulated.

5.5 Societal and practical impact

In the DeFi market, vetting a counterparty is expensive, and the capacity to do it is unevenly spread. Wealthy actors may be able to pay for proprietary risk tools, while retail participants work from anecdotal evidence or skip vetting altogether. A score built from the permissions users have already granted, and cheap to recompute (Table 4.1), lowers the barrier to vetting the contracts that ask for those permissions. Such a score shows how many owners would be exposed if a contract misused its allowances, which transfer records do not show; an approval granted earlier is one route by which tokens are drained (Section 2.17). It is not shown to anticipate drains: the drain heuristic rose with every score, the transfer-based ones included (Section 4.6).

5.6 Sybil attack cost model on allowance graphs

On a transfer graph, a malicious actor can inflate in-degree and in-flow artificially by moving assets among accounts it controls, paying for this in transaction fees and possible inventory loss (Cong et al., 2023). Allowance graphs expose a different attack surface, since each edge records an authorization, not an actual transfer. In this section we formulate a simple cost model for inflating EndorseRank with fake allowances. It builds on the optimal link farms of Gyöngyi and Garcia-Molina (2005) and on the result of Cheng and Friedman (2005) that symmetric functions such as PageRank are not Sybil-proof (Section 2.6), and adds the cost of an allowance edge and the size of the gain on this study's graph.

5.6.1 Attack setup

Let the attacker control a set of addresses $\mathcal{S}$, and write $\mathcal{H} = V \setminus \mathcal{S}$ for the honest addresses in the endorsement graph $G_{\text{approve}} = (V, E, w)$. An edge (u, v) of weight $w_{uv} > 0$ records that u has

authorized v to spend its tokens. The weight is that of Equation (3.7). With damping parameter $d \in (0, 1)$, EndorseRank finds the PageRank vector $\mathbf{r}$ on G_{approve} as the solution of (Page et al., 1999):

$$\mathbf{r} = d\, P^{\top}\mathbf{r} + \frac{1-d}{|V|}\mathbf{1}, \tag{5.2}$$

where P is the transition matrix of Equation (2.1). A node with out-edges splits its row in proportion to its outgoing allowance weights. A dangling node, one without out-edges, has every entry of its row equal to $1/|V|$, so its mass is spread uniformly over all nodes (Langville & Meyer, 2006). The fixed point is the same as that of the iteration in Equation (3.1), which the implementation runs and which feeds dangling mass back through the uniform teleportation vector. The attacker's objective is to maximize the mean EndorseRank of a target set $\mathcal{G} \subseteq \mathcal{S}$,

$$J(\mathcal{G}) = \frac{1}{|\mathcal{G}|}\sum_{i \in \mathcal{G}} r_i, \tag{5.3}$$

subject to a budget constraint on the cost of the attack, which is defined next.

5.6.2 Cost components

The attacker creates a set of edges $E_{\mathcal{S}}$, all with sources in $\mathcal{S}$, and may also buy a set $E_{\mathcal{H}\to\mathcal{S}}$ of edges from honest owners into $\mathcal{S}$. Costs are counted over q observation windows, the number of score refreshes for which the targets are meant to hold their rank. Gas is the only cost that every edge incurs.

1. Gas cost. Each `approve` or `permit` call consumes gas; write c_{gas} for the expected fee of one allowance update. EndorseRank reads the `Approval` events from 1 October 2023 to the anchor in every graph here, and an allowance stays until it is revoked or replaced, so an approval emitted once enters every later graph; inside a closed farm its time weight does not matter, since row normalization removes it (item 2). A closed farm therefore pays its gas once, and only a score over a rolling window of fixed length would make it emit its approvals

again:

$$C_{\text{gas}} = |E_{\mathcal{S}}|\, c_{\text{gas}}.$$

2. Exposure cost. An approval locks no tokens, and the owner need not hold any balance of the token when it approves. The amount costs nothing either: an unlimited allowance takes the full value weight V without any balance behind it, and after row normalization a single out-edge gives $P_{uv} = 1$ whatever its weight. An edge inside $\mathcal{S}$ therefore exposes nothing and costs nothing beyond gas. Exposure arises only when an attacker address that holds tokens approves a spender $v \notin \mathcal{S}$, for instance to look like an ordinary user of known contracts, since v can then move those tokens. Such an edge also lets score mass leave $\mathcal{S}$. With ℓ_{uv} the expected cost to the attacker of one such exposure over one window,

$$C_{\text{exp}} = q \sum_{(u,v)\in E_{\mathcal{S}},\ v\notin\mathcal{S}} \ell_{uv},$$

which is zero for a closed farm.

3. Revocation and detection cost. Legitimate users revoke old allowances from time to time, and a farm of fresh addresses that approve only one another might be identifiable. Let c_{rev} be the expected cost per window of detection and of the revocations the attacker issues to avoid it. Over q windows,

$$C_{\text{rev}} = q\, c_{\text{rev}}.$$

4. Bought honest edges. The attacker can also pay honest owners to approve its addresses, for instance through an app that asks users for permission. A bought edge (h, a), with $h \in \mathcal{H}$ and $a \in \mathcal{S}$, stays in the graph while h leaves it in force but loses share of h's row as it ages, so keeping it effective costs, per window,

$$c_{\text{in}}(h, a) = c_{\text{gas}} + p_h + \rho_h, \tag{5.4}$$

where p_h is the payment to the honest owner and ρ_h is a risk premium the owner asks against the allowance being spent by the attacker.

Over the q windows, the total cost of the attack is

$$C = C_{\text{gas}} + C_{\text{exp}} + C_{\text{rev}} + q \sum_{(h,a)\in E_{\mathcal{H}\to\mathcal{S}}} c_{\text{in}}(h, a). \tag{5.5}$$

For a closed farm, $C_{\text{exp}} = 0$ and $E_{\mathcal{H}\to\mathcal{S}}$ is empty, so $C = |E_{\mathcal{S}}|\, c_{\text{gas}} + q\, c_{\text{rev}}$.

5.6.3 Influence and a cost-effectiveness bound

By the usual sensitivity results for PageRank, an edge moves $\mathbf{r}$ in proportion to the stationary mass at its source node and to its normalized outgoing weight (Langville & Meyer, 2006). Consider first a closed farm: every node of $\mathcal{S}$ has at least one out-edge, all of these edges end in $\mathcal{S}$, and no honest owner approves an address in $\mathcal{S}$. Let D be the set of dangling nodes, all of which then lie in $\mathcal{H}$, and let $r_D = \sum_{j\in D} r_j$ be their stationary mass. By Equation (5.2), a node $i \in \mathcal{S}$ receives $d \sum_{j\in\mathcal{S}} r_j P_{ji}$ from the farm, $(1-d)/|V|$ from teleportation and $d\, r_D/|V|$ from the dangling rows. The rows of the farm's nodes sum to one inside $\mathcal{S}$, so summing over $\mathcal{S}$ gives $\sum_{i\in\mathcal{S}} r_i = d \sum_{i\in\mathcal{S}} r_i + |\mathcal{S}|\,(1 - d + d\, r_D)/|V|$, that is,

$$\sum_{i\in\mathcal{S}} r_i = \frac{|\mathcal{S}|}{|V|} \cdot \frac{1 - d + d\, r_D}{1 - d}. \tag{5.6}$$

The first factor is the farm's population share, and the second says how many times that share the farm collects. The second factor equals one only when $r_D = 0$, that is, on a graph without dangling nodes; otherwise the dangling rows pass part of the honest mass to the farm. Since the dangling nodes lie outside $\mathcal{S}$, $r_D \le 1 - \sum_{i\in\mathcal{S}} r_i$, and substituting this into Equation (5.6) gives a bound that holds on any graph:

$$\sum_{i\in\mathcal{S}} r_i \le \frac{|\mathcal{S}|}{(1-d)\,|V| + d\,|\mathcal{S}|}. \tag{5.7}$$

For a farm that is small against $|V|$, the bound is close to $1/(1-d)$ times the population share. On the EndorseRank graph of the matched cohort at T_{obs}, with $6{,}219{,}063$ nodes, the pipeline's own

PageRank code gives $r_D = 0.355$, 0.382 and 0.404 at $d = 0.75$, 0.85 and 0.95, so a closed farm on this graph collects 2.06, 3.17 and 8.68 times its population share. A direct check with the same code agrees: a ring of 20 addresses, each approving the next, added to the graph collects 3.17 times its population share at $d = 0.85$.

Equation (5.7) limits the farm's total mass, not the score of one target, and the attacker can concentrate that mass. Take a star in which f farm addresses each approve the target t and t approves each of them back. Every farm address a has t as its only out-neighbor, so $P_{at} = 1$, and the row of t sums to one over the farm. Write $c = (1 - d + d\, r_D)/|V|$ for the mass that each node receives from teleportation and the dangling rows. The balance equations $r_a = c + d\, r_t P_{ta}$ and $r_t = c + d \sum_a r_a$ then give

$$r_t = \frac{c\,(1 + df)}{1 - d^2}.$$

At $d = 0.85$, $c = 7.64 \times 10^{-8}$, which is the common score of every address that receives no allowance. With $f = 10$ (eleven addresses and twenty approvals), the target reaches $r_t = 2.62 \times 10^{-6}$ on the graph with the star added, above the median EndorseRank of the cohort contracts but only 8.2×10^{-5} of the highest (0.0318). The score grows linearly in f: each further farm address adds $d\, c/(1 - d^2)$ to r_t and costs two approvals. Solved for f, the formula needs about 22 farm addresses to reach the cohort's 90th percentile, about 2,074 its 99th and about 135,895 its highest score. Looking ordinary therefore takes a few dozen approvals and reaching the top about 270,000, two per farm address, since the highest-ranked cohort contract holds about 3% of the graph's stationary mass. This study did not measure c_{gas}, so these counts are not turned into a price.

Honest in-edges are therefore not needed to reach the upper part of the ranking, but bought ones add to the gain. When the farm has no out-edges to honest addresses, an edge (h, a) bought from an honest owner h adds, to first order,

$$\Delta \sum_{i \in \mathcal{S}} r_i \approx \frac{d\, r_h\, P_{ha}}{1 - d}$$

to the farm's mass, where P_{ha} is the share of h's row that the new edge takes. A fresh large or unlimited approval bought from h, with $V(z)$ close to one, takes about $\sigma(0)/(o(h) + \sigma(0))$ of h's row; for the median owner in the graph that share is 0.957. The edge an attacker would buy is therefore a large or unlimited approval from an owner whose own approvals are old, small or few, preferably one with a high r_h. Such an approval puts the owner's balance of the token at the attacker's disposal, so the risk premium ρ_h in Equation (5.4) grows with what the owner holds and is close to zero only for an owner that holds none of the token. In the capped allowance layer of C-PR such an approval takes a large share of a row only if it is large against the owner's other allowances (Section 3.6.3). The attack's cost-effectiveness is

$$\eta = \frac{\Delta J(\mathcal{G})}{C}, \tag{5.8}$$

with $\Delta J(\mathcal{G})$ the gain in the objective of Equation (5.3) that the attack edges produce, held over the q windows that C in Equation (5.5) pays for.

Equations (5.6)–(5.8) show that EndorseRank, as defined here, is open to cheap inflation by closed farms. Under uniform teleportation and uniform redistribution of dangling mass, a closed cluster is not held to its population share. The control that does bound closed clusters changes the walk itself: if teleportation and dangling redistribution go only to a vetted seed set, as in TrustRank (Gyöngyi et al., 2004), a cluster outside that set gains mass only through edges from honest owners, which the attacker has to buy at the price of Equation (5.4). AWP's restart rule gives the transfer side no protection either: a farm whose addresses send small amounts to one another raises their activeness and so draws restart mass to itself, and EndorseRank with activity restarts inherits the same exposure. Under C-PR, which restarts uniformly, a farm whose addresses only approve follows the allowance layer alone, so Equation (5.6) applies with the coupled graph's dangling mass. The Sybil-adjusted stability proxies (Section 3.9.3) flag repeated transfers from a few addresses but not a farm of fresh addresses, so they cannot replace an adversarial assessment, which we leave to future work.

5.7 Regulatory and ethical considerations

On-chain reputation metrics raise questions of transparency as well as equity. Earlier legal work has shown that decentralized finance markets remain bound by disclosure and consumer protection rules even though settlement happens on-chain (Zetzsche et al., 2020). Packin and Lev-Aretz (2024) add a warning of their own: decentralized credit scoring may perpetuate black-box harms when models cannot be interpreted or contested, or when they are linked to identity databases without due process. Both EndorseRank and AWP are computed deterministically from on-chain events and a public daily price series with open-source algorithms, so anyone can recalculate them (Appendix 8.1). That openness partly answers the black-box criticism. It leaves the distributional concern untouched, because contracts with little activity, new ones among them, will score lower whatever their code. Ethically, these scores should never be used as security ratings, as creditworthiness measures in the traditional sense, or as evidence that the owners who approve a contract are distinct individuals (Douceur, 2002). They are rank-based indicators of a contract's position in the defined graph topology. If a deployment grants financial services on the strength of score thresholds, it should clearly document the graph definitions behind the scores, the timeframe, and the limitations, above all the exposure to Sybil attacks. A regulatory framework centered on interpretability would prefer this kind of ranking to proprietary machine learning models like those Kotb (2023) reviews, as long as providers make no exaggerated claims about how well they can predict loan repayment rates (Packin & Lev-Aretz, 2024).

Privacy on an open blockchain is limited by the ledger's very nature (Nakamoto, 2008), and computing the reputation scores draws on nothing beyond public data. Further attempts to uncover the identity behind each address are discouraged, and the scores should not be treated as permanent, fingerprint-like identifiers. This study scores contract addresses alone and makes no attempt to link any address to a real-world individual (Section 3.16).

5.8 Threats to validity

The findings reported here should be read against the limitations below, which are grouped under construct, internal, external and adversarial validity.

5.8.1 Construct validity

The threats below concern whether the scores and checks measure what they are read as measuring.

- These ratings are not credit ratings: no default labels from uncollateralized lending enter the analysis.
- Same-window checks rest on local allowance and transfer statistics.
- Holdout construct. The holdout labels belong to the constructs of the scores (Section 5.1.4), so neither is an independent credit or trading construct. The approval label also tracks short-term continuation: in a check made after the labels were known, outside every rule, the number of owners whose latest approval to a spender was at most 30 days old at the freeze ranks new approval pairs at 0.579 $[0.564, 0.596]$ in W0 and 0.606 $[0.587, 0.623]$ in W1, above every score, although 86% and 89% of the spenders have no such owner (folder `experiments/exploratory/2026-10-14-endorsement-count-variants`), so the raw degree is not the strongest simple baseline.
- Unit of analysis. EndorseRank scores the receiving side of an approval, and every scored address here is a contract (Section 4.1.1). Claims about the reputation of traders rest on the W0 trader run and the comparison of Section 5.1.6, both on small samples.
- Own-construct agreement is partly mechanical. Each score smooths over the whole graph the in-degree of its own edges, so it is expected by construction to agree with its own family; the within-score contrasts of Table 8.6 compare like with like but remain checks of the instruments.
- Value weights. EndorseRank and AWP pass each US-dollar amount through $V(z)$ with a slope set so that the median amount weighs one half (Section 3.5.1). The slopes are choices of the re-

vision, checked only in the same window (Table 4.7). The layers of C-PR add dollar amounts without the bound of V, capping each allowance at U (Section 3.6.3). On both spender holdouts EndorseRank ranks both labels slightly better than the allowance layer walked alone (Tables 4.10 and 4.16), without a paired contrast, so how amounts are weighted may matter for this construct, and neither weighting is shown to be the right one.

5.8.2 Internal validity

The threats below concern whether the comparisons are free of choices that could have produced their results.

- Shared observation window. Every method and proxy is computed over 1 October 2023 to 30 June 2026. Market regimes peculiar to those months (volatility, incentive programs, protocol changes) may push alignment up or down. The checks on damping and value slopes measure sensitivity to these parameters, but none of them varies the calendar window.
- Cohort selection. The matched cohort is the 15,107 contracts on Arbitrum One that received non-zero approvals of one of the five tokens from at least three owners in the window, and results may not carry over to contracts with fewer approvals or with approvals of other tokens only. Nor do the graphs see every cohort contract (Section 4.1.1), and the sample-definition sensitivity shows how strongly the coefficients react when the evaluation population changes (Table 4.8).
- Baseline fidelity. AWP follows the published method in its edge weights and its restarts (Section 3.6.1). The paper gives no parameter values, so k, t_0 and b are this study's choices (see value weights above). The paper scored Ethereum transactions; here AWP scores transfers of five ERC-20 tokens on Arbitrum One, valued in US dollars, on the cohort's ego network, which limits the activeness of a sender outside the cohort to its transfers to cohort contracts.
- Order of analysis. The cohorts, scores, labels, statistics and rule A were fixed in the analysis plan of 9 October 2026, before any log of the study was extracted. Rule B with its margin,

the two sensitivity analyses and the comparison on trader outcomes were registered later that day, after the W0 results were known and before the study extracted the July–September logs (Section 3.7.2). Every other comparison, the degree contrasts among them, was computed after the relevant labels were known and is reported with intervals; none of them decides a question.

- Dependence and leakage. Contracts that share owners or a protocol are resampled as independent, and the post hoc contrasts are not adjusted for multiplicity (Section 3.10); the W0 data were extracted around a cohort defined with the approvals of the W0 label window (Section 3.7).
- Revision of the amounts. The amounts were revised on 10 October 2026, after the registered results of all three windows were known (Section 3.5.1). The revised analysis, the main analysis of this thesis, is therefore post hoc, and its verdicts under rules A, B and R1–R3 are a re-analysis, not registered tests. The revision was committed to the public repository before any revised score was computed, its slopes and cap come from quantiles of the events up to t_1, and the registered analysis reached the same verdicts (Appendix 8.6). Some contrasts differ between the two, among them the sign of Q in W1, F3 (Table 8.12) and S-PR against the transfer walk on new senders in W0.
- Timing environment. All runtimes are wall-clock times from one machine (Appendix 8.1). The standard deviations in Table 4.1 cover only repeated solves of one configuration, and in the damping sweep each graph–damping pair is timed once. Variation between machines is not measured, so the comparison is stated in ratios instead of seconds.

5.8.3 External validity

The threats below concern how far the results carry beyond the data studied.

- Single chain and window. The evidence comes from Arbitrum One alone, from contracts that at least three owners approved on one of five tokens, over 33 months of scored activity

and two three-month label windows. Extending the findings to other rollups, Layer-1 chains or lending-centered venues would need cross-protocol validation, since DeFi venues differ in protocol design (Werner et al., 2022) and in market structure (Schär, 2021).

- Sample scaling versus alignment scaling. The subsample stages bear only on runtime on subgraphs of the matched-cohort graph (Section 4.2.1); they do not show that the τ profile holds at other population sizes.
- Ego-network depth. Each graph keeps only the edges incident to the cohort contracts, so the findings about propagation do not carry over to graphs with longer endorsement paths (Section 5.3.2).

5.8.4 Adversarial validity

The threats below concern what an attacker could do to the scores.

- Sybil and wash trading. Rankings may be manipulated through adversarial allowance or transfer patterns (Douceur, 2002), and the cost model of Section 5.6 is analytical, not an empirical red-team evaluation.
- Strategic approval behavior. Once EndorseRank is live, rational actors may grant allowances strategically to farm rank. An observational τ estimated before deployment may overstate how well the ranking resists such behavior.

5.9 Limitations

On this sample, EndorseRank's strengths are agreement with the allowance checks, a ranking distinct from AWP's, a low cost owed to graph sparsity and an order for the spender contracts without a transfer edge, about one in six, which walks over transfers score zero (Section 5.1.4). Its weaknesses are a ranking that carries little transfer information, a holdout coefficient below that of the raw in-approve degree it smooths in both windows (computed after the labels were known), exposure to closed farms below the top of its ranking, and nothing to say about contracts that few owners

approve or about externally owned accounts, which are not scored (Section 3.2).

The coupled operator was tried at three fixed values of λ, with uniform teleportation and a per-node convex mixture at the transition level, on two layers weighted by dollar amounts, with each allowance capped at U. Layers weighted as in EndorseRank and AWP, with activity restarts, were coupled only after the labels were known (C-AWP, Section 4.6.4); node-dependent or learned weights, mixtures of the stationary vectors instead of the transition rows, and other Markov chains went untested. Both answers to RQ5 therefore apply only to the coupling the plan fixed, on this dataset.

The data add four limits. The graphs hold five tokens only (Table 8.4), so a contract that owners approve only on other tokens is outside the cohort. Every price comes from one public service, the daily series of DefiLlama (DefiLlama, n.d.), so an error in it on some date would pass into every amount of that day (Appendix 8.4.2). The account type was read once for each address, at blocks of 9 October 2026 after both label windows (Table 8.1), so a contract that destroyed itself before EIP-6780 (Ballet et al., 2023) is left out as an externally owned account (Section 4.1.1). And the trader cohorts are small (Section 4.1.1), so every trader result has wide intervals.

EndorseRank draws on no off-chain evidence about who controls an address; its only off-chain inputs are the token list and the daily price series. Unlike attestation tooling such as Human Passport, which uses identity checks, biometrics and vouching to establish that an address belongs to a distinct person (Human Passport, n.d.), it gives no guarantee that two highly ranked addresses are controlled by different actors, so it should be read as an ordering of an already admitted population and not as a Sybil defense. Counting only the approvals of owners that pass an attestation gate before computing EndorseRank is a plausible deployment, but the gate's own weaknesses would carry over to it (Bordeianu & Popescu, 2026), and evaluating the combination needs labeled Sybil data that this study did not collect.

For this cohort of contracts and time frame, EndorseRank is a cheap, allowance-specific reputation score for spender contracts; it is not a credit scoring method, nor a Sybil-resistant identity mechanism, nor a substitute for protocol-level collateralization (Gudgeon, Perez, et al., 2020). The allowance edge it rests on is worth adding to on-chain reputation, and the approval count is the

cheapest way to use it.